HOTEL BOOKING ANALYSIS

Data Analysis Internship | CyberSNova

Analysis Report

1. Executive Summary

This project analyzes 119,390 hotel booking records to understand booking cancellations and identify patterns that may affect occupancy and revenue. Overall, 44,224 bookings were cancelled (37.0%), while 75,166 were not cancelled. Cancellation rates differ by hotel type, deposit policy, lead time, market segment, and customer type. These findings can help the hotel chain improve booking policies, forecast demand, and target customer-retention efforts.

2. Business Problem

The hotel chain has reported increasing booking cancellations, which can create revenue uncertainty, reduce occupancy, and make room allocation more difficult. The purpose of this analysis is to identify patterns associated with cancellations and recommend practical actions to reduce avoidable cancellations and improve room utilization.

3. Objectives

- Measure the overall booking cancellation rate.
- Compare cancellation rates across hotel types, deposit types, market segments, and customer types.
- Examine the relationship between lead time and cancellation behavior.
- Use visualizations and a dashboard to communicate findings clearly.
- Recommend data-informed actions to improve occupancy planning and customer retention.

4. Dataset Description

The supplied Hotel Booking dataset contains 119,390 rows and 32 columns. Relevant fields include hotel, is_canceled, lead_time, arrival date fields, market_segment, deposit_type, customer_type, adr (average daily rate), total_of_special_requests, reservation_status, and reservation_status_date. The target field is is_canceled, where 1 indicates a cancelled booking and 0 indicates a booking that was not cancelled.

Data quality note: the dataset includes missing values in some fields, including agent, company, country, and children. Missing values should be handled according to the purpose of each analysis; missing agent/company values should not automatically be treated as errors because some bookings may not use an agent or company.

5. Methodology

1. Loaded the CSV dataset using pandas and reviewed its dimensions, column names, and missing values.
2. Calculated the cancellation rate from the is_canceled indicator.
3. Grouped bookings by hotel type, deposit type, market segment, customer type, and lead-time bands.
4. Compared cancellation rates across groups and created charts to communicate the patterns.
5. Interpreted results as associations rather than proof that any single factor causes cancellation.

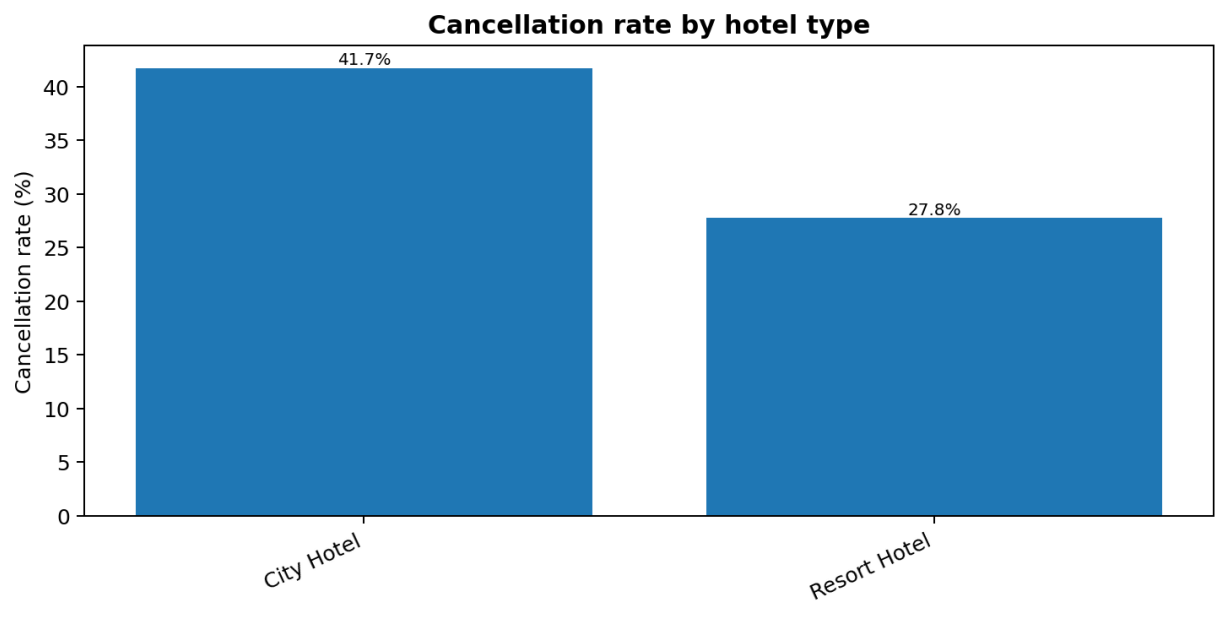
6. Key Findings

6.1 Overall cancellations

The overall cancellation rate is 37.0% (44,224 cancelled bookings out of 119,390 records). This indicates cancellations are a material operational issue and should be included in occupancy forecasting.

6.2 Hotel type

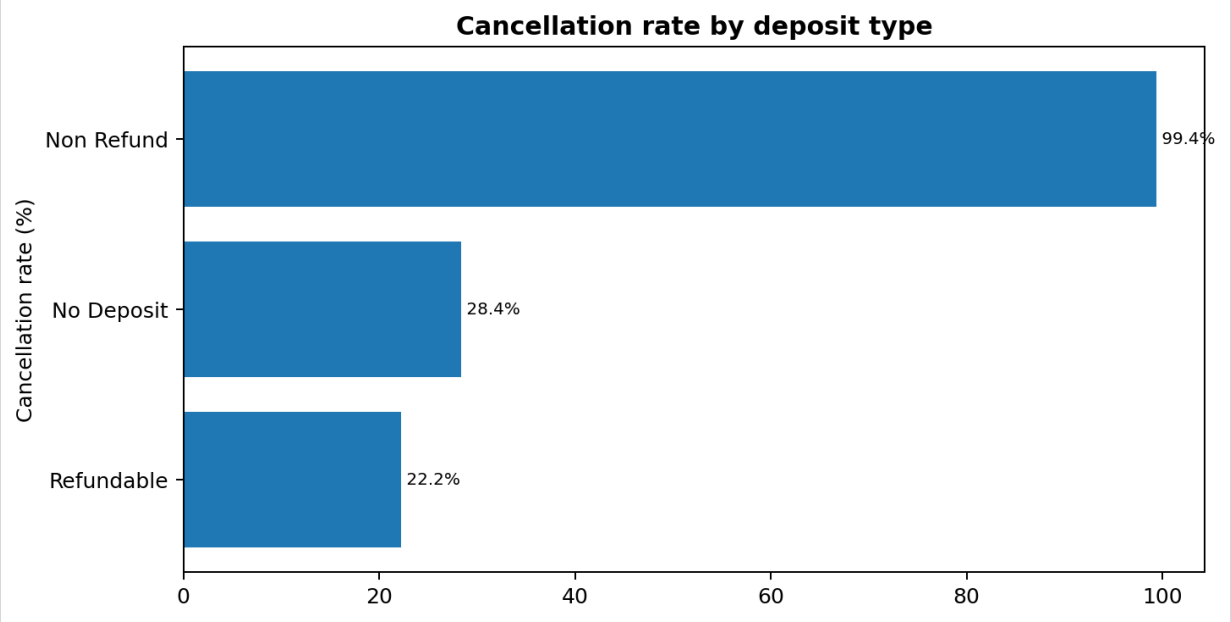
- City Hotel: 41.7% cancellation rate.
- Resort Hotel: 27.8% cancellation rate.



6.3 Deposit type

- Non Refund: 99.4% cancellation rate.
- No Deposit: 28.4% cancellation rate.
- Refundable: 22.2% cancellation rate.

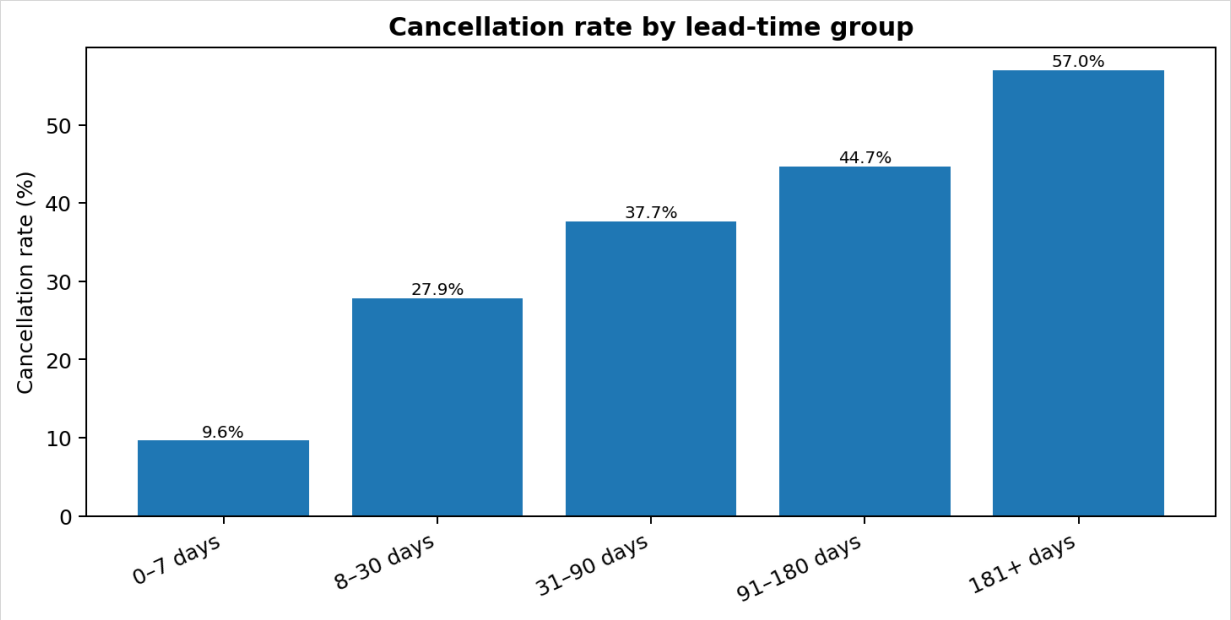
The Non Refund category has an unusually high observed cancellation rate in this dataset. This should be validated with the reservation-status and deposit-policy definitions before making policy changes; the result may reflect how this category is recorded in the source data.



6.4 Lead time

Average lead time is 144.8 days for cancelled bookings and 80.0 days for bookings that were not cancelled. This comparison suggests that bookings made further in advance may warrant closer monitoring, but lead time alone does not explain every cancellation.

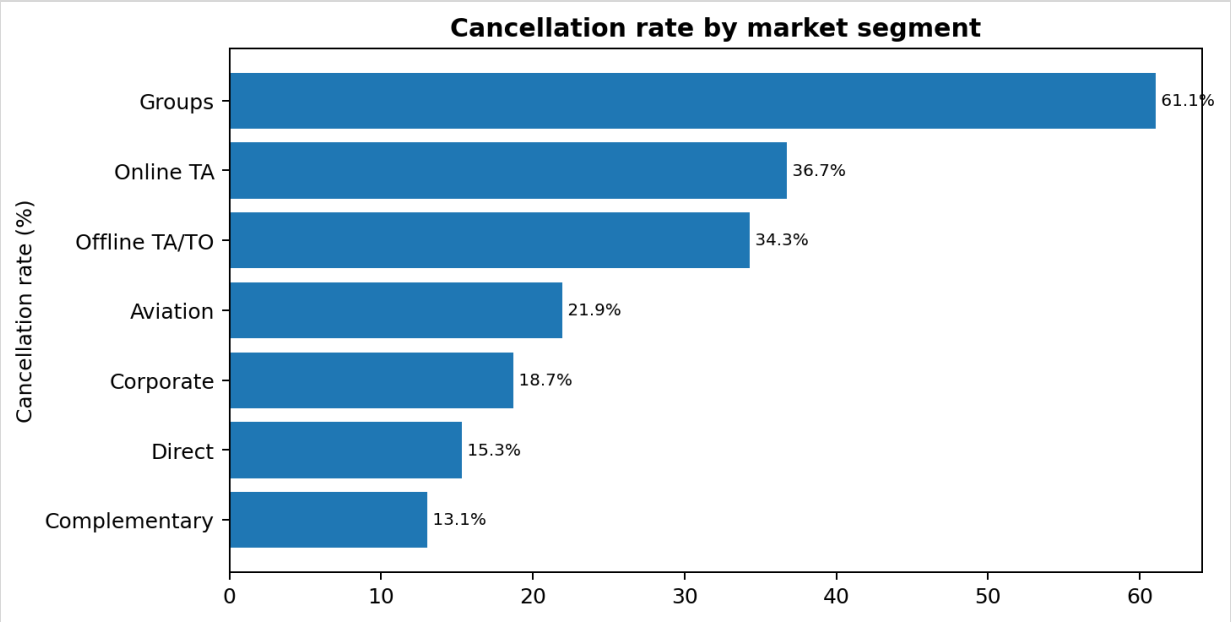
- 0–7 days: 9.6% cancellation rate.
- 8–30 days: 27.9% cancellation rate.
- 31–90 days: 37.7% cancellation rate.
- 91–180 days: 44.7% cancellation rate.
- 181+ days: 57.0% cancellation rate.



6.5 Market segment

- Undefined: 100.0% cancellation rate.
- Groups: 61.1% cancellation rate.
- Online TA: 36.7% cancellation rate.

- Offline TA/TO: 34.3% cancellation rate.
- Aviation: 21.9% cancellation rate.
- Corporate: 18.7% cancellation rate.
- Direct: 15.3% cancellation rate.
- Complementary: 13.1% cancellation rate.



6.6 Customer type and special requests

- Transient: 40.7% cancellation rate.
- Contract: 31.0% cancellation rate.
- Transient-Party: 25.4% cancellation rate.
- Group: 10.2% cancellation rate.

Cancellation rates also vary by the number of special requests. This can be explored in the dashboard to understand whether engagement signals are associated with a higher likelihood of completing a booking. Such relationships should be interpreted as patterns, not causes.

7. Recommendations

6. **Improve cancellation forecasting:** Use lead time, hotel type, deposit type, market segment, and customer type as monitoring variables in weekly occupancy forecasts.
7. **Review deposit-policy data:** Investigate the unusually high cancellation rate in the Non Refund category and verify business rules and data definitions before changing deposit policies.
8. **Focus on high-cancellation segments:** Review booking terms and communication for segments with higher cancellation rates. Consider clearer cancellation deadlines, confirmation messages, and targeted retention offers where financially appropriate.
9. **Strengthen pre-arrival communication:** Send timely booking confirmations and reminders, especially for reservations made well in advance.
10. **Use controlled overbooking carefully:** If operational policy permits, estimate expected cancellations by hotel and booking characteristics before setting any overbooking buffer; avoid a single blanket rate.
11. **Monitor performance regularly:** Track cancellation rate, confirmed bookings, lead-time mix, and occupancy by hotel type and month in the dashboard.
12. **Validate before acting:** Test recommendations on a pilot basis and compare results over time; cancellation patterns alone do not establish causation.

8. Dashboard and Screenshots

The interactive Streamlit dashboard is intended to provide filters and visual summaries for hotel type, year, customer type, market segment, deposit type, booking status, and lead time. Capture screenshots only after confirming the dashboard loads and all charts display correctly.

- Screenshot 1: full dashboard overview showing title, filters, KPI cards, and main charts.
- Screenshot 2: cancellation rate by hotel type and deposit type.
- Screenshot 3: cancellation trend over time and lead-time analysis.
- Screenshot 4: market-segment analysis and business insights/recommendations.

9. Limitations

- The analysis identifies associations and does not prove causation.
- The dataset does not directly provide every business cost, profit, or room-availability measure needed to quantify exact revenue lost from cancellations.
- Some fields contain missing values and should be treated according to their meaning.
- The Non Refund deposit category result is unusual and should be checked against the dataset definitions before operational decisions are made.

10. Conclusion

The dataset shows an overall cancellation rate of 37.0%, with clear differences across hotel type, deposit type, market segment, and customer type. The hotel chain should combine cancellation monitoring with better pre-arrival communication, segment-aware policies, and improved forecasting. The dashboard can support ongoing monitoring, while any policy changes should be validated through further investigation and controlled testing.

Appendix: Dataset fields used

hotel; is_canceled; lead_time; arrival date fields; market_segment; deposit_type; customer_type; adr; total_of_special_requests; reservation_status.

Prepared for the CyberSNova Data Analysis Internship. Replace or add your name and internship details before submission if required.